Testing the Longevity Bottleneck Hypothesis

Genomics of Rejuvenation and Ageing Lab, University of Birmingham

Charlie Hinson

The dromaeosaur Saurornitholestes predating on an early mammal. Creative Commons https://w.wiki/EwuN

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Rejuvenomics
Genomics of Ageing and Rejuvenation Lab

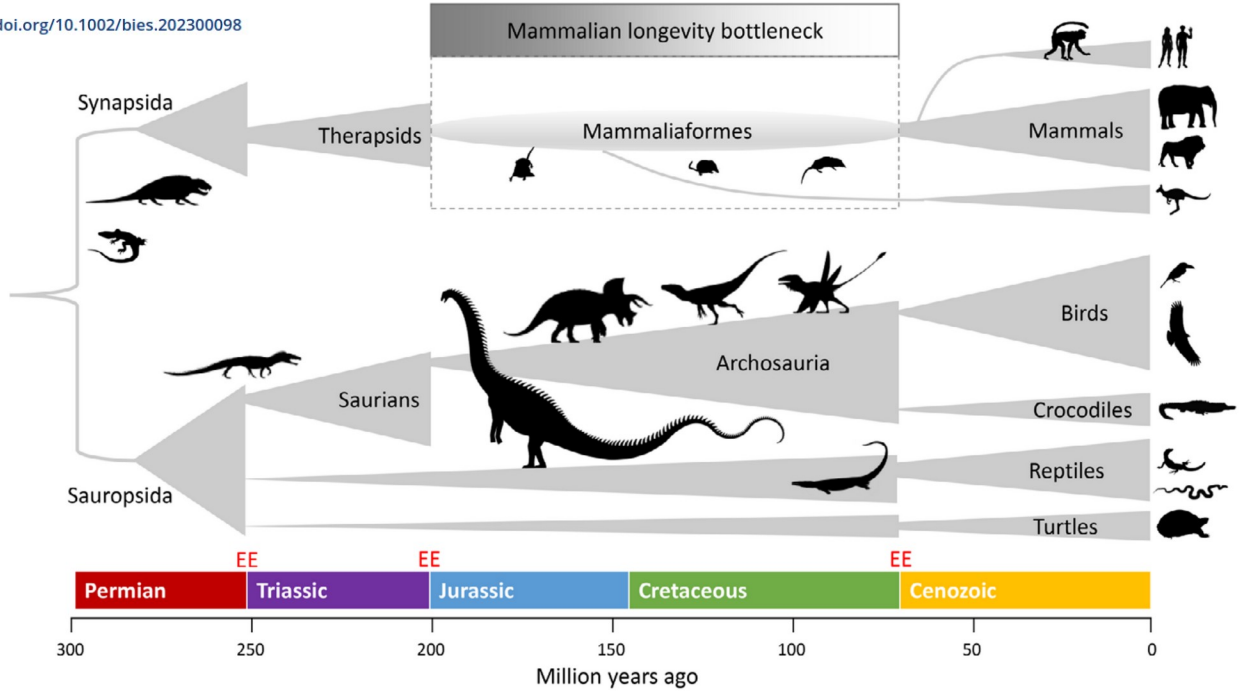
What is the Longevity Bottleneck Hypothesis (LBH)?

IDEAS & SPECULATIONS | Open Access

The longevity bottleneck hypothesis: Could dinosaurs have shaped ageing in present-day mammals?

João Pedro de Magalhães

First published: 28 November 2023 | https://doi.org/10.1002/bies.202300098



Steps for testing

- Compare life histories of organisms tested via AnAge
- dN/dS testing of gene orthologs relating to ageing via Ensembl, SNAP, NCBI, MEGA, R
- Following

A beginners guide to estimating the non-synonymous to synonymous rate ratio of all protein-coding genes in a genome

Daniel C Jeffares [1], Bartłomiej Tomiczek, Victor Sojo, Mario dos Reis

Affiliations + expand

PMID: 25388108 DOI: 10.1007/978-1-4939-1438-8_4

NCBI ⇨ MEGA Molecular Evolutionary Genetics Analysis ⇨ R

Ensembl

dN/dS Ratios

- **dN**: non-synonymous changed per non-synonymous site
- **dS**: synonymous changes per synonymous site
- As synonymous mutations accumulate at a constant rate under neutral selection, a gene's dN/dS ratio can tell us the proportional selective pressure upon a gene relative to a neutral background mutation rate at the synonymous sites.

Fossil evidence of mammalian presence in dinosaur diets

Articles

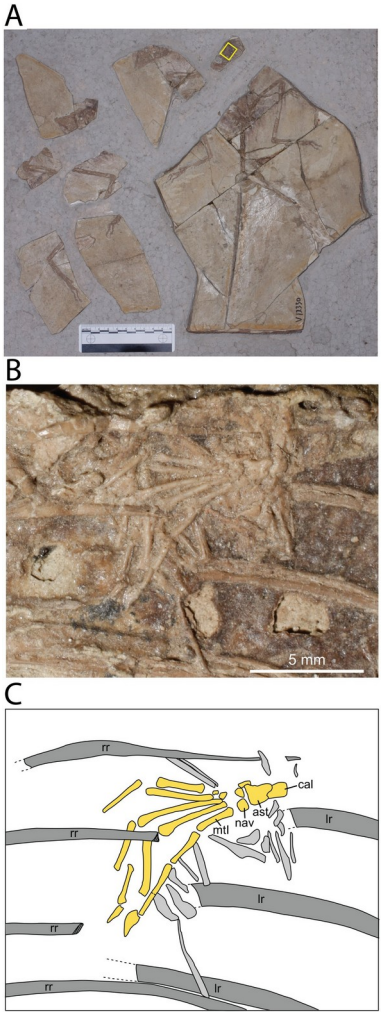
Generalist diet of *Microraptor zhaoianus* included mammals

David W. E. Hone, T. Alexander Dececchi, Corwin Sullivan, Xu Xing & Hans C. E. Larsson

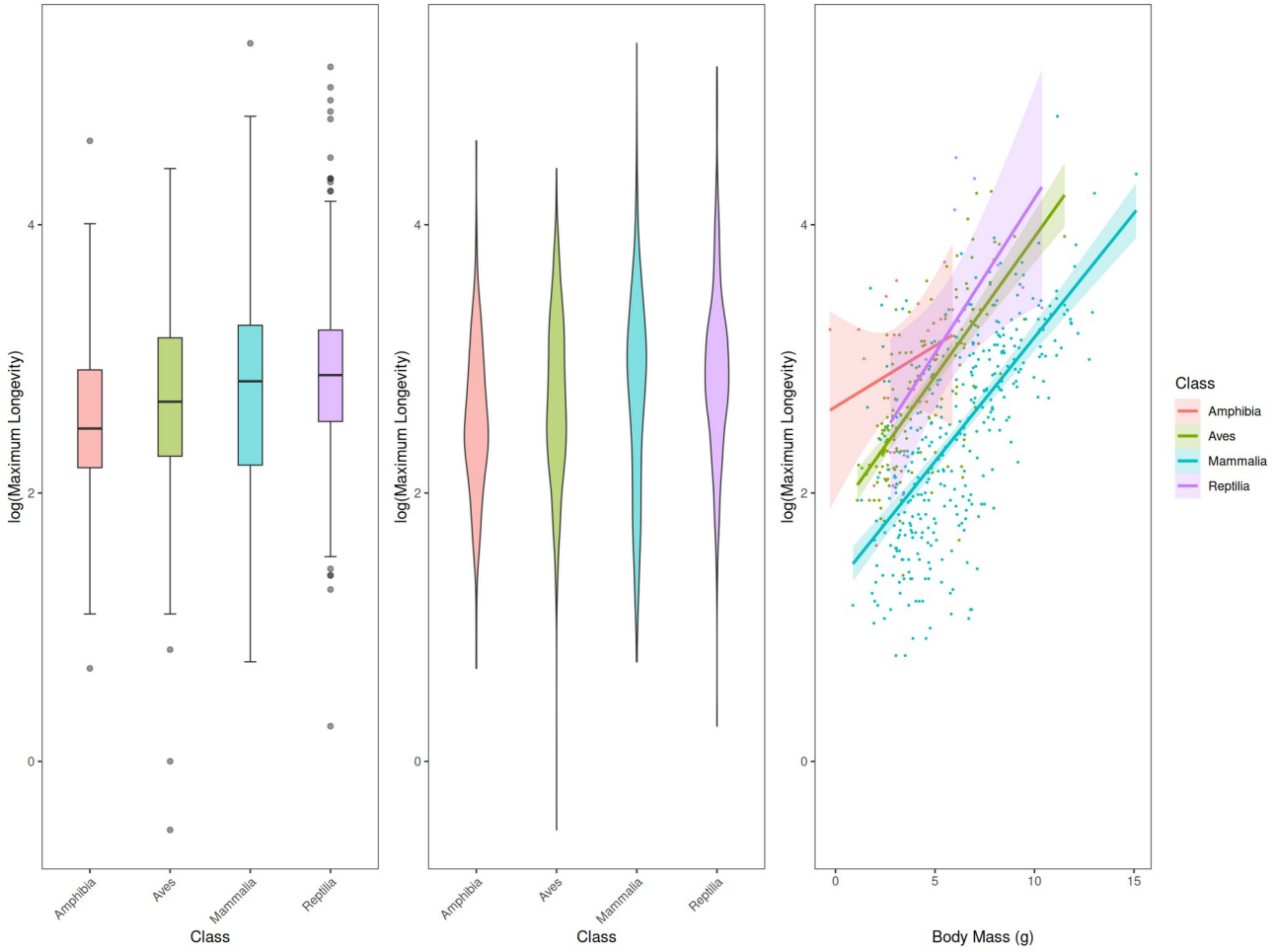
Article: e2144337 | Received 05 Oct 2021, Accepted 27 Sep 2022, Published online: 20 Dec 2022

Cite this article https://doi.org/10.1080/02724634.2022.2144337 Check for updates

Microraptor zhaoianus, Creative Commons https://w.wiki/EwuN



Brief overview of terrestrial vertebrate longevity

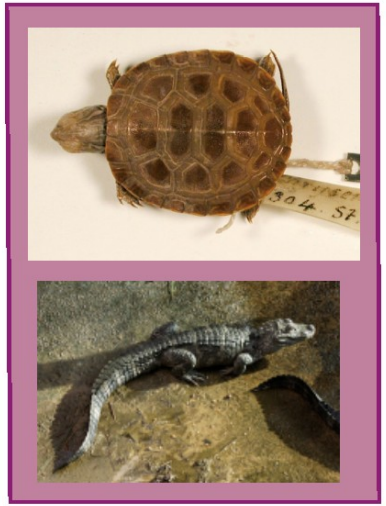


Genes investigated

Pathway/gene	Further Information
Photolyase/UV-repair	Light-dependant DNA repair mechanism. Found in most other organisms, including vertebrates, with the exception of mammals. Orthologs (cryptochromes) act as a control for assessed out-groups.
IGF-1	Insulin growth factor 1, involved in the GH/IGF-1/insulin pathway. Heavily influences vertebrate development. Downregulation purported to be beneficial towards human lifespan. (Vitale et al. 2019)
mTOR	Mechanistic target of rapamycin, an extensively studied universal regulator of cell proliferation and metabolism. (Papadopoli et al. 2019)
PARP-1	Poly (ADP-ribose) polymerase coding genes involved in DNA repair pathways. Also may interact with the protein causing Werner syndrome in humans (Bürkle et al. 2005).

Representative animals used in dN/dS analysis

- Monotremes: Platypus, echidna, diverged ~180mya (Zhou et al. 2021)
- Marsupials: Opossum, tammar wallaby, diverged ~160 mya (Luo et al. 2011)
- Eutherian basal lineages: Tenrec, hedgehog, tree shrew
- Out-groups to compare: Green anole, painted turtle, Chinese alligator, axolotl, Western clawed frog



Images courtesy of the Smithsonian Institute

Method

Distance Estimation

Option	Setting
ANALYSIS	
Scope	*Pairs of taxa*
ESTIMATE VARIANCE	
Variance Estimation Method →	*Bootstrap method*
Bootstrap Replicates →	*1000*
SUBSTITUTION MODEL	
Substitutions Type →	*Syn-Nonsynonymous*
Genetic Code Table →	*Standard*
Model/Method →	*Nei-Gojobori method (Proportion)*
Substitutions to Include →	*Difference (n - s)*
RATES AND PATTERNS	
Rates among Sites →	*Uniform Rates*
Pattern among Lineages →	*Same (Homogeneous)*
DATA SUBSET TO USE	
Gaps/Missing Data →	Pairwise deletion

M12: Analysis Preferences

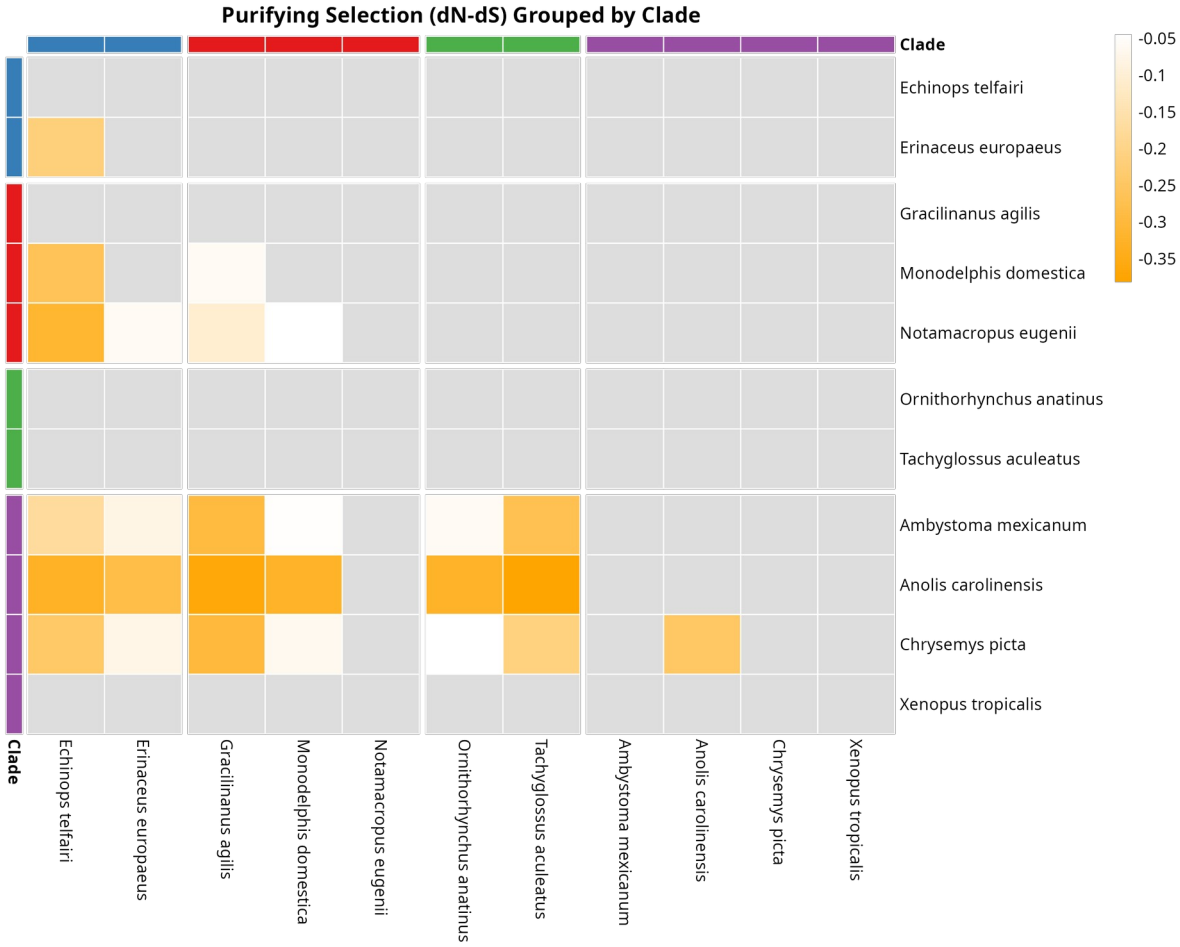
Z-test of Selection

Option	Setting
ANALYSIS	
Scope	*In Sequence Pairs*
Test Hypothesis (HA: alternative) →	*Positive selection (HA: dN > dS)*
ESTIMATE VARIANCE	
Variance Estimation Method →	*Analytical method*
SUBSTITUTION MODEL	
Substitutions Type →	*Syn-Nonsynonymous*
Model/Method →	*Nei-Gojobori method (Proportion)*
DATA SUBSET TO USE	
Gaps/Missing Data →	Pairwise deletion

- Test for positive selection, look for significance
- If none, test for purifying selection, look for significance
- Find pairwise distance of dN - dS

NCBI ⇨ MEGA Molecular Evolutionary Genetics Analysis ⇨ R

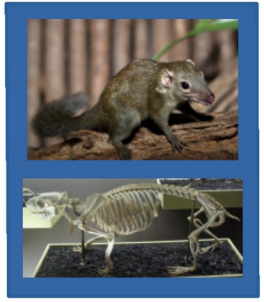
Results: IGF-1



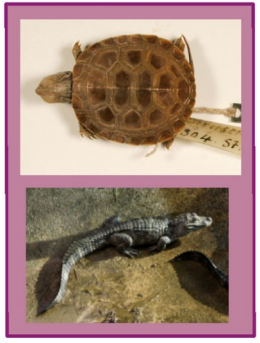
Monotremes

Marsupial

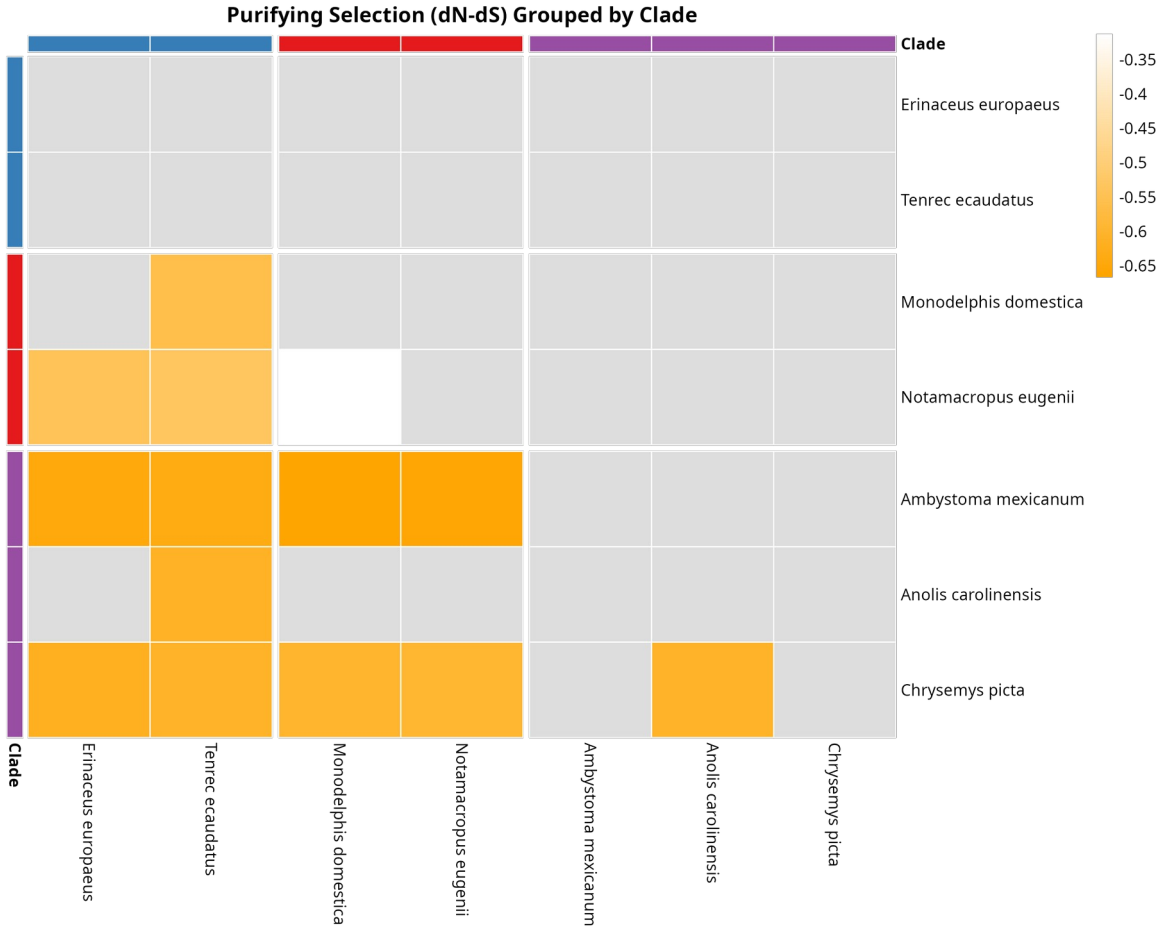
Eutherian



Out-groups



Results: mTOR

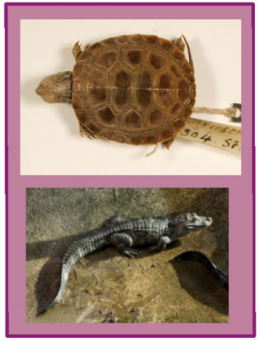


Monotremes

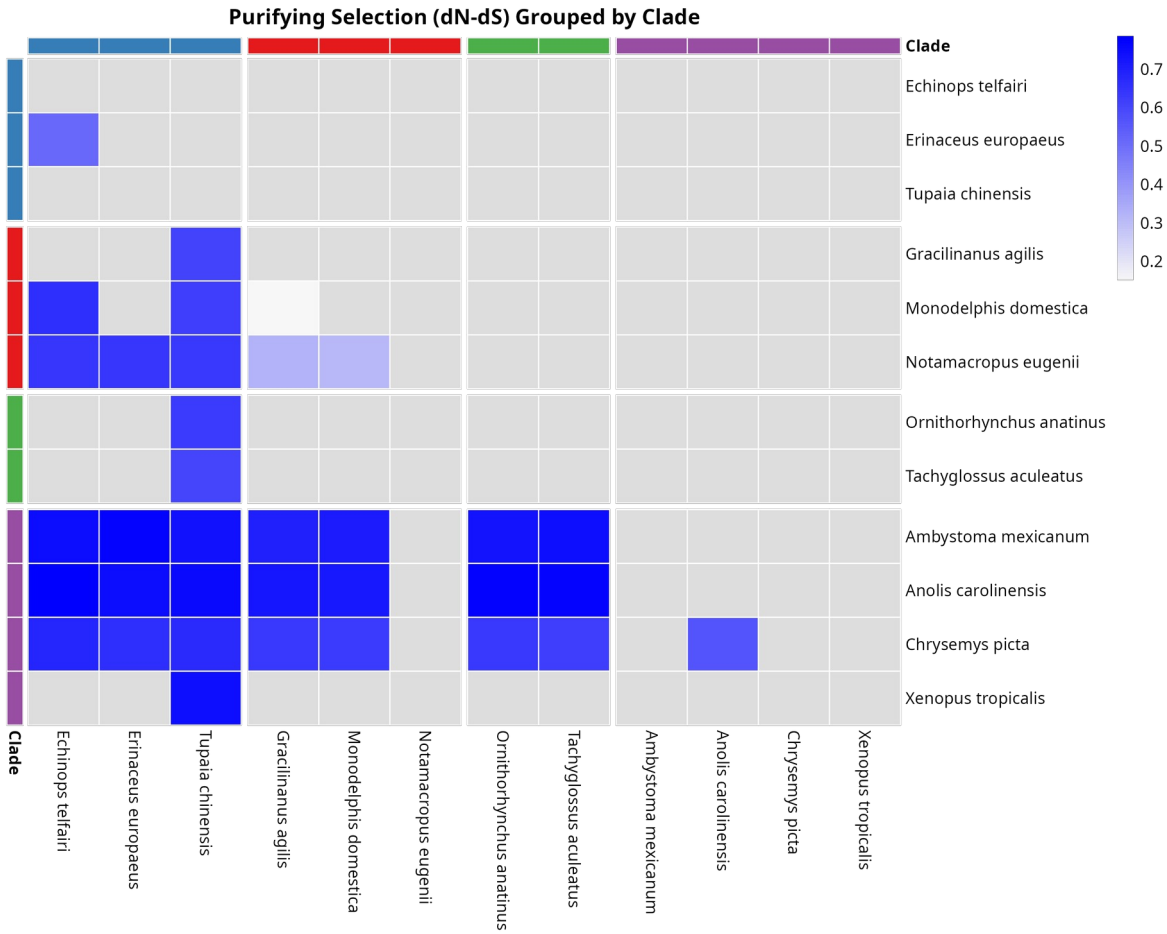
Marsupial

Eutherian

Out-groups



Results: PARP-1

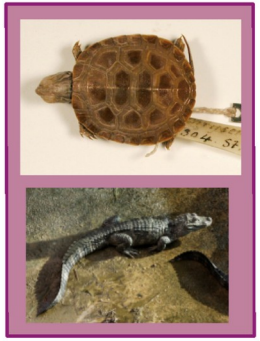


Monotremes

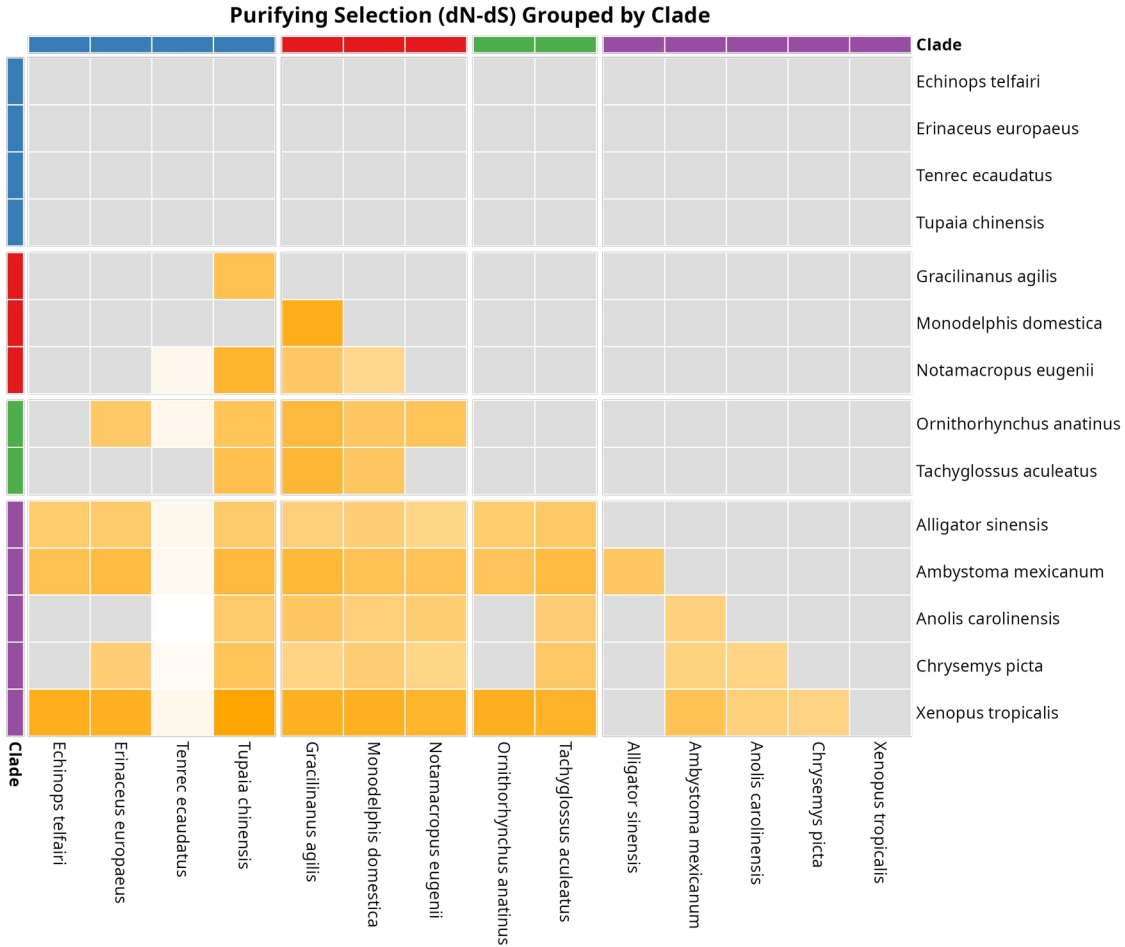
Marsupial

Eutherian

Out-groups



Results: Photolyase/Cryptochromes (Control)

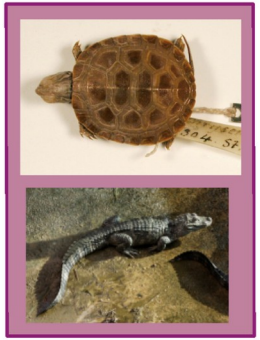


Monotremes

Marsupial

Eutherian

Out-groups



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